Unnamed_Unit

Unit 3 - Algorithm Writing

PDF Documents

PDF Document 1: 1738215972265-Chapter 9 - Unit 3 - Algorithm Writing.pdf

This PDF document is included in the unit materials.

UNIT 3 ALGORITHM WRITING

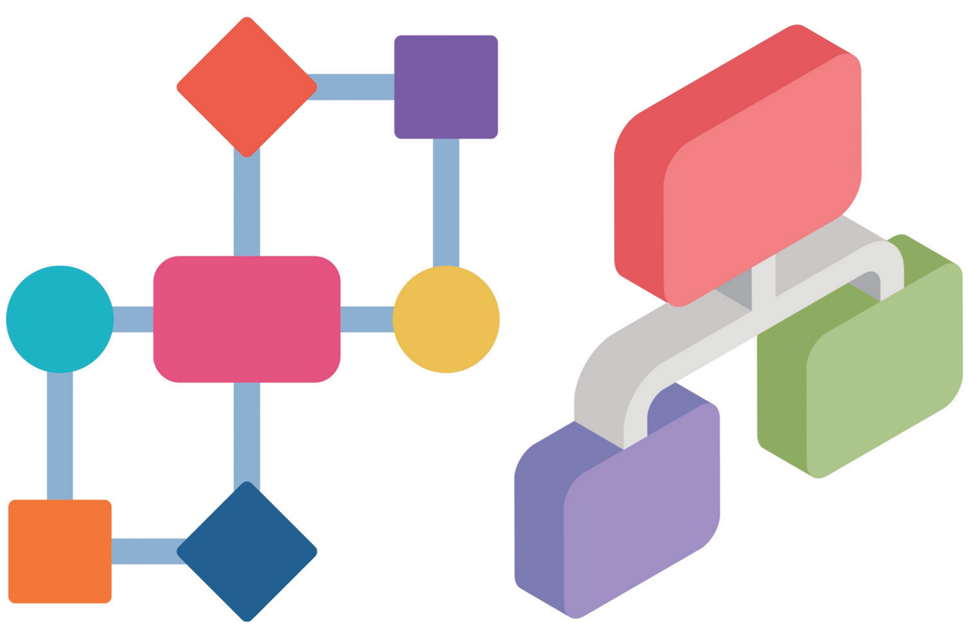
Unit Introduction

In Unit 3, you will focus on writing algorithms for your smart device project. Algorithms are step-by-step instructions that tell your device how to perform specific tasks. By creating clear and precise algorithms, you ensure that your device operates smoothly and efficiently. This unit will guide you through the process of translating your ideas into detailed algorithms, setting the foundation for flowchart development in the next unit.

WHAT IS ALGORITHM?

An algorithm is a special set of rules or instructions that tell you how to do something. It's like a recipe, but for solving problems or completing tasks.

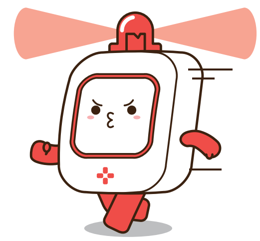
Imagine you want to make a peanut butter and jelly sandwich. The recipe with all its steps is like an algorithm for making a sandwich. It tells you exactly what to do, one step at a time, until you have a finished sandwich.

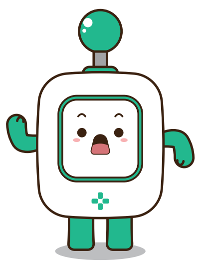


Developing Algorithm

TRANSLATING IDEAS TO ALGORITHMS

Use the ideas you developed in Unit 2 as a starting point. For each idea, write a corresponding algorithm. Be detailed and specific, explaining each action and decision your device needs to make.



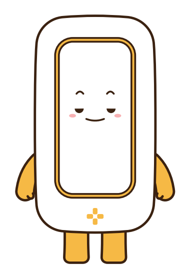


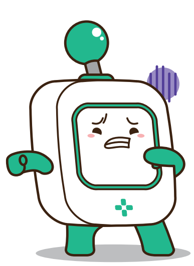
DETAILING THE STEPS

For each step in your algorithm, specify the inputs, processes, and outputs. Clearly define what each MODI module will do at every stage. This detail will help you later when you start creating flowcharts and coding your device.

STRUCTURING YOUR ALGORITHMS

Organize your algorithms logically. Break down complex tasks into smaller, manageable steps. Ensure that each part of the algorithm flows smoothly into the next, creating a clear and logical progression.





REVIEWING AND REFINING

Review your algorithms to check for clarity and completeness. Make sure each step is easy to understand and follow. Refine your algorithms as needed to ensure they are as efficient and effective as possible.

Example

SMART PLANT WATERING SYSTEM

Goal :

Automatically water the plant when the humidity level is below a certain threshold.

LOGIC (IDEAS)

1. Start
   - Begin the process.
2. Read Sensor Data
   - Read the current humidity level from the Environment module.
   - Read the current temperature from the Environment module.
3. Display Data
   - Show the current temperature and humidity level on the Display module.
4. Check Humidity Level
   - If the humidity level is below 30%:
     - Turn on the LED to yellow.
     - Play a warning sound using the Speaker module.
     - Activate the Motor B to spray water for 5 seconds.
     - Turn off the LED.
   - Else:
     - Display "Humidity is sufficient" on the Display module.
     - Keep the LED off.
5. Loop
   - Wait for 10 minutes.
   - Go back to Step 2.
6. End
   - End the process.

LOGIC (IDEAS)

1. Start
   - Begin the process.
2. **Read Sensor Data**
   - Read the current humidity level from the Environment module.
   - Read the current temperature from the Environment module.
3. **Display Data**
   - Show the current temperature and humidity level on the Display module.
4. **Check Humidity Level**
   - If the humidity level is below 30%:
     - Turn on the LED to yellow.
     - Play a warning sound using the Speaker module.
     - Activate the Motor B to spray water for 5 seconds.
     - Turn off the LED.
   - Else:
     - Display "Humidity is sufficient" on the Display module.
     - Keep the LED off.
5. **Loop**
   - Wait for 10 minutes.
   - Go back to Step 2.
6. **End**
   - End the process.

ALGORITHM

Step 1:
- The algorithm starts.

Step 2:
- The system reads the current humidity and temperature from the Environment module.

Step 3:
- The current temperature and humidity levels are displayed on the Display module.

Step 4:
- The system checks if the humidity level is below 30%.
- If it is, the LED turns yellow, a warning sound is played, and the Motor B sprays water for 5 seconds.
- If the humidity is sufficient, a message is displayed, and the LED remains off.

Step 5:
- The system waits for 10 minutes before reading the sensor data again, creating a loop.

Step 6:
- The algorithm ends.

Translating Ideas to Algorithms

TRANSLATING IDEAS TO ALGORITHMS

Use the ideas you developed in Unit 2 as a starting point. For each idea, write a corresponding algorithm. Be detailed and specific, explaining each action and decision your device needs to make. Look at the example in the previous page, and translate your ideas to algorithm.

INSTRUCTIONS

- **Start with Your Ideas:** Use the ideas you came up with in Unit 2. Think about what your smart device needs to do.
- **Write Detailed Steps:** For each idea, write down detailed and specific steps (algorithm) that your device needs to follow. Explain every action and decision.
- **Use the Example:** Look at the example on the previous page to help you understand how to translate your ideas into an algorithm.

Example

LOGIC (IDEAS)

1. Start
2. Measure soil moisture
3. If moisture is below a certain level,
4. Display the status
5. Water the plant until the humidity reaches the desired level

TRANSLATION FOR MODI MODULES (ALGORITHM)

1. System Starts.
2. If the Environment module detects humidity below 60%,
3. The Display module shows the humidity level.
4. Set Motor A to 60°.
5. Maintain the set angle until the humidity level reaches 75%.

Be in the MODI modules' shoes! By imagining yourself as part of the system, you will easily translate your ideas into the machine's thinking processes.

(Continue...)

ALGORITHM LOGIC (IDEAS)	TRANSLATION FOR MODI MODULES

Structuring Your Algorithms

Organize your algorithms logically. Break down complex tasks into smaller, manageable steps. Ensure that each part of the algorithm flows smoothly into the next, creating a clear and logical progression.

INSTRUCTIONS

Put Yourself in the MODI Modules' Shoes:

- Imagine you are one of the MODI modules to see if there are any missing steps in your algorithm.

Break Down the Steps:

- Think about how you can break down the steps you wrote before into smaller, more detailed steps.

Refine Your Algorithm:

- Write down the big steps you wrote earlier and break them down into smaller, more detailed steps.

Example

YOUR TRANSLATION	REFINED ALGORITHM
1. System Starts.	1. System Starts.
2. If the Environment module detects humidity below 60%,	2. If the Environment module detects humidity below 60%,
3. The Display module shows the humidity level. 4. Set Motor A to 60°.	3. Shows the humidity level on the Display module and Set Motor A to 60°.
5. Maintain the set angle until the humidity level reaches 75%.	4. Maintain the set angle of Motor A until the humidity level reaches 75%. 5. If the humidity level reaches 75%, set Motor A to the original angle

Exercise

Now, it's your turn to refine your previous algorithm!

YOUR TRANSLATION	TRANSLATION FOR MODI MODULES

Detailing the Steps & Reviewing and Refining

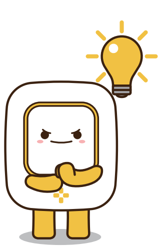
This time, we are going to detail the steps (our last check) and review the entire algorithm to ensure the flow is well-structured.

DETAILING THE STEPS

For each step in your algorithm, specify the inputs, processes, and outputs. Clearly define what each MODI module will do at every stage. This detail will help you later when you start creating flowcharts and coding your device.

REVIEWING AND REFINING

Review your algorithms to check for clarity and completeness. Make sure each step is easy to understand and follow. Refine your algorithms as needed to ensure they are as efficient and effective as possible.



Read your refined algorithm you wrote in the previous unit. Is there any missing step? If there isn't, you are ready to draw a flowchart based on the algorithm! Let's move onto the next step, “flowchart”!